

**DAMILOLA ADEOYE (11918968): STATEMENT OF PURPOSE FOR ADMISSION INTO THE CHEMISTRY
DOCTORAL PROGRAM UNIVERSITY OF ALABAMA FOR FALL 2018.**

Inhibition of cell wall synthesis by attacking Penicillin Binding Proteins (PBPs) is the mode of action of most antibiotics. The discovery of penicillin brought a great succor especially during World War II as it became the antibiotic of choice in the management of a broad spectrum of infections in the 1940's. However, microbes, capable of fighting off the slightest form of attack threw out an arsenal of resistance against the new wonder drug. Hence, in a matter of time, penicillin like the sulfonamides that predates it had to battle the evolution of antibiotics resistance. Over the years, we have seen the introduction of several other antibiotics from **Streptomycin to Chloramphenicol, Tetracycline, Erythromycin, Vancomycin, Methicillin, Ampicillin and Cephalosporins**. Yet, we have had to battle the same problem of resistance against each molecule. In the light of this, we moved towards drug combinations especially for "Hospital Acquired Infections" for synergistic activity. This has led to the use of **Ampicillin & Cloxacillin, Amoxycillin & Clavulanate Potassium, Sulfamethoxazole & Trimetoprim** etc. in the treatment of diseases. The sad news however is that treatment cost of ailments skyrockets while microbes acquire more virulent and resistant powers. The approach to microbial infection treatment in the years gone has been one of two; "cidal" or "static". We left microbes with two options. To die (instantly or gradually) or fight back. They chose the latter and that is why we are where we are today.

Many different approaches to curbing microbial resistance has been mooted in scientific literatures. They include; **the inhibition of toxin function, targeting microbe toxin function delivery, targeting the regulation of virulence expression in microbes, inhibition of microbe adhesion to host**, etc. These approaches are not direct attacks on microbes but a form of protection of host from microbial toxins. Hence, the bacterial need not to fight back as has been the case with current resistance. It's worthy to note that these approaches are still subject to research and it's with this in mind that I find the Chemistry Graduate Program at the University of Alabama as a very attractive place to pursue my interest. Dr. Matthew Thompson's research effort in finding "New Strategies to Combat Fosfomycin Resistance" is a good match for my interest. His approach to this research has been centered on identifying the cause of the resistance to come principally from the fosfomycin modifying enzyme **FosB** of the **Vicinal Oxygen Chelate (VOC)** superfamily common to all drug-resistant pathogens. I am enthralled to know that the next goal in this effort is in the identification

of molecules that will suppress and inhibit this enzyme. While I can't really categorize his approach to dealing with Antibiotic Resistance based on the approaches earlier stated, I am confident that his approach presents a lot of therapeutic opportunities on which I can lay the foundation for an outstanding career in researches that would be renowned for healthcare innovations and revolution in the area of infectious disease management and immunity. I am certain that answers to this research problems would be found using tools in enzymology, bioinformatics, molecular biology, etc. Fortunately, my research interests encloses all these areas and I am looking forward to this new challenge.

I graduated with a first degree in chemistry (top 10%) from one of Nigeria's leading Universities, University of Ilorin, I proceeded to Nigeria's premier University, University of Ibadan where I bagged a degree in Analytic Chemistry, finishing in the top two in my class. That degree in analytical Chemistry exposed me to a host of analytical techniques ranging from spectroscopic, radiochemical and several biological assay approaches relevant for this pursuit. I decided to pursue the master's degree in preparation for a career in industrial quality control/assurance which I was very passionate about prior to graduate studies at University of Ibadan. This passion was fueled by my experiences during internships at the "Quality Assurance/Quality Control" (QA/QC) department of a FMCG manufacturing company and a Pharmaceutical firm in Nigeria were my first real life practical experiences about implications of results of chemical analysis on events in everyday life. With every samples that were analyzed in the laboratory, our (chemists) approval or otherwise determined if a batch will move up in the manufacturing chain or not. This realization of the power a chemist can wield outside the walls of a laboratory after each analysis was a defining moment as par which career path to tow after graduation and it influenced my decision to undertake a Masters' degree in Analytical chemistry. At that point in time, I felt such degree will strongly prepare me for a lifetime career in Industrial QA/QC.

However, course work, research and interdisciplinary interactions in graduate school exposed me to the enormous potentials embedded in Analytical Chemistry and its applicability in fields ranging from Environmental monitoring, molecular biology, enzymology, hydrobiology, entomology, agronomy, food technology, etc. My degrees in Chemistry and Analytical Chemistry provides a dynamic platform through which I can navigate into order disciplines with little or no limitations. The game-changer was however the induced scarcity of tomato fruits in Nigeria in 2016 caused by

a moth, *Tuta absoluta* (A.K.A. Tomato Ebola) utterly changed my perspective and initiated me into the world of chemical-biology. That experience was life changing for me in that I had never felt the impact of a commodity's scarcity prior to that experience. With price rising by over 400%, it affected my feeding pattern as a student. As a result of that challenge, I downloaded loads of literature and immersed myself in a deep study of the processes leading to fruit spoilage and possible ways to mitigate them in a sustainable way. I realized that most fruit spoilage are consequences of fungal attack on fruit cell walls through the secretion of cell wall degrading enzymes like xylamase, polygalacturonase, pectinase, hemicellulase, etc. Initially, I imagined how a simple but reversible kinetic reaction between an inhibitor and the enzyme responsible for fruit spoilage can easily be initiated to delay this process using the Lock and Key enzyme models but over time, I have come to recognize that the solutions to this problem is as likely to be rooted in genomics as it is in enzymology. This was how I discovered chemical biology. Fortunately, fungi and bacteria infections are as common in humans as they are in plants are animals. This makes a pursuit of this interest more satisfying for me.

My research experience is quite diverse. I take this to be an advantage in the pursuit of this objective. Research for me as always been an interesting journey especially when I think about my expectations of having neat experiments with straightforward results within few minutes of laboratory work. Sadly, on most days, a few minutes set-up became ten hours and sometimes, with confusing results as the end product. I found actual research to be different from my prior education. Particularly challenging was the application of knowledge gained in course works like spectroscopy. There is, however, no greater mollification than completing an experiment with good results. In college, I carried out a research that incorporated organic synthesis, organometallic synthesis and spectroscopic techniques. At the end of that research, my team successfully synthesized and characterized a Schiff base (an imine) derived from the condensation of Isatin (a carbonyl) and aniline (an amine). The Schiff base was subsequently complexed with dibutyl-tin (II) Chloride to form an organotin compound which has potential therapeutic properties. Results of spectroscopic and C.H.N. analysis of the compounds confirmed our preliminary expectations. Graduate school research was quite different as it was focused more on quantitative research due to the fact that I had chosen to specialize in Analytical chemistry. Working with Professor Percy Onianwa as a research student during grad school research project was quite different and much more demanding than I experienced with Dr. Owolude. I had to get used to working independently but intelligently and with little supervision. The research objective was to evaluate the

water quality and heavy metal pollution indices of water bodies in Ibadan, Africa's largest indigenous city. It was a totally different experience for me in that it involved a lot of field work that entails a detailed design of sampling plans, transportation, budgeting, classical and numerous instrumental analysis techniques. However, I embraced the challenges with a positive attitude, leading a research team of seven to finish the research in record time. At the end of the year, Prof rated me as the best of his seven graduate students.

My experiences are however not limited to the walls of laboratories or academic institutions. In the past, I have taken up leadership roles in the administration of community development groups that labored assiduously to ensure that the education and healthcare needs of rural dwellers and the underprivileged in the society are well catered for. I took up active roles such as "Financial Secretary" and "Training Secretary" in the "Education Development Group" and also, "Transport and Organizing Secretary" at the "Nigeria Christian Corpers' Fellowship" during the compulsory one year national youth service in my country. I have also taught and mentored research students in the past (See resume for details).

I currently work with one of Nigeria's leading Pharmaceutical companies and my job as a Medical representative makes me exposed to the array of challenges needing urgent attention in the walls of hospitals that only researches in Chemical-Biology labs can resolve. We would one day need alternatives to Cefuroxime, Ceftriaxone, Amoxyl-Clav and all in the fight against microbes. The days are fast coming upon us. It is with these days in mind that I put up my application for PhD admission into the Chemistry Doctoral Program at the University of Alabama, Tuscaloosa for fall 2018. Having had many correspondences the Director of Graduate Recruiting of the Chemistry Program, Dr. Marco Bonizzoni, I have come to the conclusion that the department is a good environment to pursue my interest. I hope that my background and research aspirations would be enough to land this life changing opportunity.